DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW

SESSION- 2024-25 PERIODIC ASSESSMENT – II

SET -II

MD. Hifzaan Khan

SUBJECT- MATHEMATICS (041)

SAMPLE QUESTION PAPER

M. M: 80

Time: 3 hrs.

Class IX Sec D

Name: 6/9/24

General Instructions:

- *All questions are compulsory*
- *Section A: Question 1-18 are objective type and conceptual based and question 19 and 20 are assertion and reason type each consist of 1 mark each.*
- *Section B: Question 21-25 consist of 2 marks each.*
- *Section C: Question 26-31 consist of 3 marks each.*
- *Section D: Question 32-35 consist of 5 marks each.*
- *Section E: Question 36-38 consist of 4 marks each containing two case study based questions with subparts of 1+1+2 marks.*
- *Internal choices are provided in each section. Use of calculator is not permitted.*
- *This question paper consists of 6 pages.*

SECTION A

Directions: Question no. 1 to 18 are MCQ / Conceptual approach and question 19 and 20 are assertion and reason type each consist of 1 mark.

1. The product of any two irrational numbers is 1
   a) Sometimes rational, sometimes irrational
   b) Always a rational number
   c) Always an integer
   d) Always an irrational number.

2. In a histogram the class intervals or the groups are taken along 1
   a) Y – axis b) X - axis
   c) Both of X - axis and Y - axis d) none of these

3. If $\left(\frac{3}{4}\right)^6 \times \left(\frac{16}{9}\right)^5 = \left(\frac{4}{3}\right)^{x+2}$, then the value of x is 1
   (a) 2 (b) 4 (c) -2 (d) 6

   $a^3+b^3 = (a+b)(a^2-ab+b^2)$

4. If $\frac{a}{b}+\frac{b}{a}=1, (a\neq 0, b\neq 0)$, then the value of a^3+b^3 is 1
   (a) –1 (b) 0 (c) 1 (d) $\frac{1}{2}$

   $a^2+b^2 = ab$

   $a^3-b^3 = (a-b)(a^2+b^2+ab)$

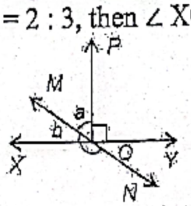
   $= (a-b)(ab+ab)$

   $= (a-b)(2ab)$

   $= \cdot$

5. Any point on the x – axis is of the form 1
   (a) (x, y) (b) (0, y) (c) (x, 0) (d) (x, x)

6. In the given figure (not drawn to scale), lines XY and MN intersect at O. If $\angle POY = 90°$ and a : b = 2 : 3, then $\angle XON$ is equal to _____. 1



   a) 180° b) 130° c) 126° d) 90°

7. The value of n for which $\sqrt{n}$ be a rational number is 1
   (a) 2 (b) 4 (c) 3 (d) 5

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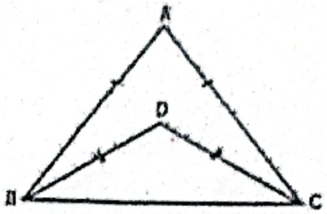
8. The distance of the (4, –3) from x – axis is 1
(a) 3 units (b) –3 units (c) 4 units (d) 5 units

9. If a wheel has six spokes equally spaced, then the measure of the angle between two adjacent spokes is 1
(a) 90^0 (b) 30^0 (c) 60^0 (d) 180^0

10. The number of rational numbers between $\sqrt{3}$ and $\sqrt{5}$ is 1
(a) One (b) 3 (c) none (d) infinitely many

11. The value of $(32)^{\frac{1}{5}} + (-7)^0 + (64)^{\frac{1}{3}}$ is 1
a) 10 b) 0 c) 1 d) 11

12. In the given figure, the ratio ∠ABD : ∠ACD is 1
(a) 1 : 1 (b) 2 : 1 (c) 1 : 2 (d) 2 : 3



13. The figure obtained by plotting the points (2, 3), (–2, 3),(–2, –3) and (2, –3) is a 1
(a) trapezium (b) rectangle (c) square (d) rhombus

14. If $(x^{10} + 2x^9 + k)$ is divisible by (x + 1) then the value of k is 1
a) 2 b) -2 c) 1 d) -3

15. Which of the following is not true for a parallelogram? 1
(a) opposite sides are equal (b) opposite angles are bisected by the diagonals
(c) opposite angles are equal (d) diagonals bisect each other.

16. The quadrilateral formed by joining the midpoints of the sides of a quadrilateral PQRS, taken in order, is a rectangle, if 1
(a) PQRS is a rectangle (b) PQRS is an parallelogram
(c) diagonals of PQRS are perpendicular (d) diagonals of PQRS are equal.

17. If one angle of a triangle is equal to the sum of other two angles, then the triangle is 1
(a) an Equilateral triangle (b) an Isosceles triangle
(c) an obtuse triangle (d) a right triangle .

18. In triangles ABC and DEF, AB = FD and∠ A = ∠ D. The two triangles will be congruent by SAS axiom if 1
a) BC = DE b) AC = DE c) AC = EF d) BC = EF

19. **Assertion:** every irrational number is a real number 1
Reason: there exist only 234 irrational numbers between 0 and 1
a) both Assertion and Reason are correct and Reason is the correct explanation for Assertion
b) both Assertion and Reason are correct and Reason is not the correct explanation for Assertion.
c) assertion is true but the reason is false.
d) both assertion and reason are false.

20. **Assertion (A):** If the diagonals of a parallelogram ABCD are equal, then ∠ ABC = 90 ° 1
Reason (R): If the diagonals of a parallelogram are equal, it becomes a rectangle.
a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

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SECTION B

Directions: Questions no. Q21 to Q25 are Short answer type questions of 2 marks each.

21. Show that 1.27272727..... can be expressed in the form of $\frac{p}{q}$, where p and q are integers and $q \neq 0$. 2

22. For what value of c, the linear equation $2x + cy = 8$ has equal values of x and y for its solution. 2

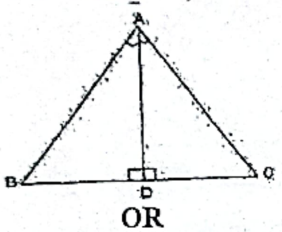
OR

Determine the point on the graph of the linear equation $2x + 5y = 19$, whose ordinate is $1\frac{1}{2}$ times its abscissa. 2

23. Find the value of a if $x + 6$ is a factor of $x^3 + 3x^2 + 4x + a$. 2

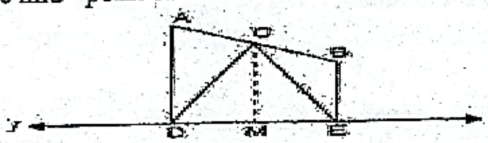
24. Diagonals of a quadrilateral ABCD bisect each other. If $\angle A = 35^0$, determine $\angle B$. 2

25. Prove that $\triangle$ ABC is an isosceles, if bisector of BAC is perpendicular to BC. 2



OR

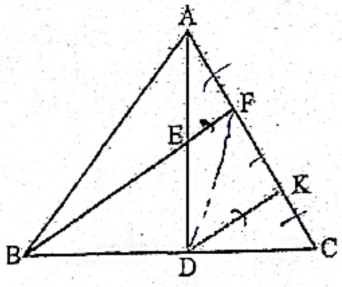
Points A and B are on the same side of a line l. AD and BE are perpendiculars to l, meeting at D and E respectively. C is the mid - point of AB. Prove that CD = CE.



SECTION C

Directions: Questions no. Q26 to Q 31 are Short answer type questions of 3 marks each.

26. In $\triangle ABC$, AD is the median through A and E is the midpoint of AD. BE produced meets AC in F such that BF || DK. Prove that $AF = \frac{1}{3}AC$ 3



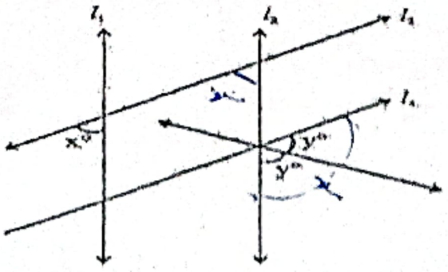
OR

In a parallelogram, the bisectors of any two consecutive angles intersects at right angle. Prove it.

27. ABC is a triangle in which $\angle B = 2\angle C$. D is a point on BC such that AD bisects $\angle BAC$ and AB = CD. Prove that $\angle BAC = 72^0$. 3

28. In figure, if $l_1 \parallel l_2$ and $l_3 \parallel l_4$. What is y in terms of x? 3

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29. The following table shows the number of scooters sold by a dealer during six consecutive years. 3

Year	2011	2012	2013	2014	2015	2016
Number of scooters sold (in thousands)	16	20	32	36	40	48

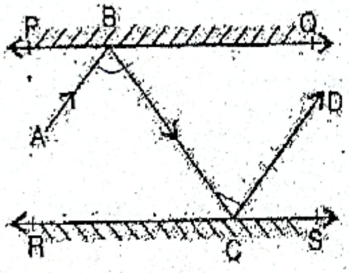
Draw a bar graph to represent this data.

30. Find the values of a and b so that the polynomial $x^4 + ax^3 - 7x^2 + 8x + b$ is exactly divisible by $(x + 2)$ as well as $(x + 3)$. 3

OR

If a, b, c are all non-zero and $a + b + c = 0$, prove that $\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab} = 3$

31. In figure, PQ and RS are two mirrors placed parallel to each other. An incident ray AB strikes the mirror PQ at B. The reflected ray moves along the path BC and strikes the mirror RS at C and again reflects back along CD. Prove that AB || CD. 3



SECTION D

Directions: Question 32-35 are long answer type questions of 5 marks each

32. The runs scored by two teams A and B on the first 60 balls in a cricket match are given below: 5
Represent the data of both the teams on the same graph by frequency polygons.

Number of balls	Team A	Team B
1-6	2	5
7-12	1	6
13-18	8	2
19-24	9	10
25-30	4	5
31-36	5	6
37-42	6	3
43-48	10	4
49-54	6	8
55-60	2	10

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33. Show that if the diagonals of a quadrilateral are equal and bisect each other at right angles, then it is a square. 5

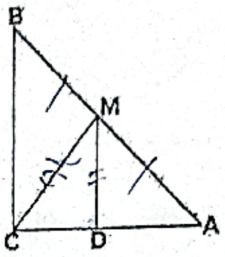
34. Represent the real number $\sqrt{2}, \sqrt{3}, \sqrt{5}$ on a single number line. (using suitable scale) 5

OR

If $x = 2+\sqrt{3}$, find (i) $x^2+\frac{1}{x^2}$ (ii) $x^4+\frac{1}{x^4}$

35. ABC is a triangle right angled at C. A line through the mid - point M of hypotenuse AB and parallel to BC intersects AC at D. show that 5

1. D is mid - point of AC
2. MD⊥ AC
3. $CM = MA = \frac{1}{2}AB$



SECTION E

Directions: Question 36-38 are case study based questions of 4 marks with subparts of 1+1+2 marks.

36. CASE STUDY - I: Collaborative Learning Can Boost Academic Success

Democracy has given people a powerful right - that is to VOTE. In India, every citizen over 18 years of age has the right to vote. Instead of enjoying it as a holiday, one must vote if he/she truly wants to contribute to the nation - building process and bring about a change.



A survey was done in a small area in which $\sqrt{9+2x}-\sqrt{2x}$ voters were men and $\sqrt{9+2x}+\sqrt{2x}$ voters were women.

Answer the following questions:-

i - Conjugate of men voters. 1

ii - at x=8 number of men voters. 1

iii total number of voters.

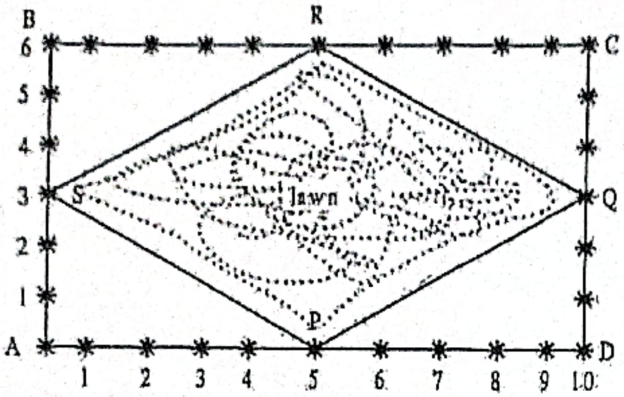
OR

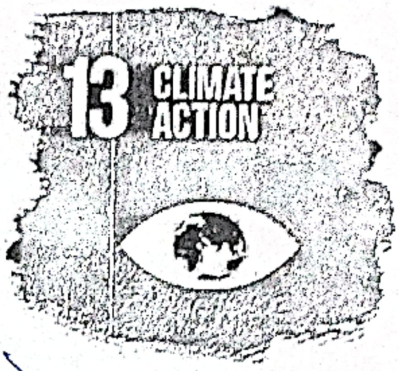
State the minimum age for voting. 2

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37. **CASE STUDY - II: Climate action @ SDG Goal under 13**

The Class IX students of a secondary school in Krishinagar have been allotted a rectangular plot of land for their gardening activity. Sapling of Gulmohar are planted on the boundary at a distance of 1m from each other. There is a lawn PQRS in the ground as shown in below figure.





(a) What are the coordinates of C, taking A as origin? (10,6) 1

(b) What are the coordinates of R, taking A as origin? (5,6) 1

(c) Find the lengths of AC and DB of the lawn. Is there any relation between them? If so, mention the property used. RMS 2

OR

Perimeter and Area of lawn are _____ 2

38.

CASE STUDY - III: Graphical Interpretation

A teacher is a person whose professional activity involves planning, organizing, and conducting group activities to develop student's knowledge, skills, and attitudes as stipulated by educational programs. Teachers may work with students as a whole class, in small groups or one - to - one, inside or outside regular classrooms. In this indicator, teachers are compared by their average age and work experience measured in years.

For the same in 2015, the following distribution of ages (in years) of primary school teachers in a district was collected to evaluate the teacher on the above - mentioned criterion.

Answer the following questions given below :-

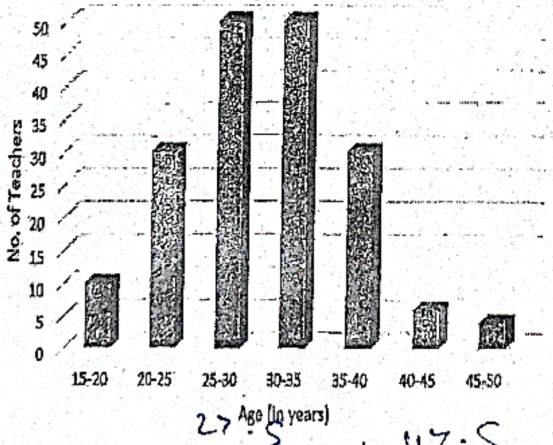
i- Number of teachers of the age group 25-30 and 40-45. 30, 5 1

ii - Compute the range of age (in years) of teachers. 1

iii -Which classes are having same no. of teachers? 25-30, 30-35 2

OR

Find the class mark of class 25-30 and 45-50? 2



27.5, 47.5

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